AYUSH RAJ

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PROFESSIONAL SUMMARY

First-year B.Tech Computer Science and Engineering (AI/ML) student at Lovely Professional University with a strong passion for technology and artificial intelligence. Built three web projects using Python, JavaScript, React, Flask and Supabase, including two hackathon projects: an outfit recommendation app and a live-deployed healthcare platform with a RAG-based AI chatbot. Seeking AI/ML, summer and winter internships, and open-source opportunities such as GSoC.

EDUCATION

Lovely Professional University, Phagwara, Punjab — Expected 2030
Bachelor of Technology in Computer Science and Engineering (Specialization: Artificial Intelligence and Machine Learning)

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, HTML5, CSS3, SQL
Frameworks and Libraries: React, Flask, Tailwind CSS, Three.js
Backend and Databases: Supabase, PostgreSQL, SQLite, REST APIs, OAuth 2.0 Authentication
AI and Automation: Machine Learning Fundamentals, RAG, LLM API Integration, Claude Code, OpenCode, n8n
Tools and Platforms: Git, GitHub, Docker, VS Code, Vercel, Linux
Design: UI/UX Design (learning), Responsive Web Design
Cybersecurity: Kali Linux, Cybersecurity Fundamentals (learning)

PROJECTS

AtmosFit: Weather-Based Outfit Recommender GitHub
Python, Flask, SQLite, JavaScript, Three.js, Open-Meteo API

- Designed outfit recommendation logic that combines feels-like temperature, rain chance, wind, UV, snow and air quality, adjusted for the user's age and gender, including children and seniors.
- Built a Flask REST API with SQLite, Google OAuth sign-in and salted password hashing, with hyperlocal weather data, a 10-day forecast and a rotatable 3D avatar (Three.js) that shows the suggested outfit.

Campus Care: LPU Healthcare Platform Live Demo | GitHub
React, Vite, Tailwind CSS, Supabase, PostgreSQL, Anthropic API, RAG

- Built a full-stack healthcare platform for LPU students and staff with health tracking, doctor and counselor appointments, video teleconsults, pharmacy ordering with hostel delivery and an interactive campus map, with separate access for students, doctors and pharmacy admins.
- Designed a Supabase backend with PostgreSQL Row Level Security, email authentication, file storage and realtime updates for doctor availability, order status and chat.
- Built an AI health assistant chatbot using Retrieval-Augmented Generation (RAG) through a Supabase Edge Function calling the Anthropic API, with a rule-based fallback, and added English, Hindi and Punjabi language support.

RailMatrix: Railway Berth and PNR Status Simulator GitHub
HTML, CSS, JavaScript (no frameworks or backend)

- Built a responsive simulator covering train search across 40 stations, class-wise availability, fares by quota and PNR status tracking, using a seeded random generator for repeatable results.
- Implemented a statistical model that predicts waitlist and RAC confirmation probability, and validated it with Node-based logic tests and end-to-end runs in headless Edge.
- Added accessibility features including keyboard-navigable tabs, ARIA labels, inline validation and a mobile-first layout tested at 360 px width.

ACHIEVEMENTS AND CERTIFICATIONS

- Participated in two hackathons as part of a four-member team.
- State-level Wushu player; gold medalist at the district-level competition, Sitamarhi, Bihar, 2019.
- AI Classroom, JioPC (certificate).
- Completed Fundamentals of Machine Learning and Artificial Intelligence, Coursera.